

vessels engaged in routine offshore logistics tasks operate fully laden with 7.4 m draft which means there will be very few occasions when the largest vessels in the industry have to make a tide dependent entry or departure through the Mermaid channel. Further the Mermaid Base will not suffer operational disadvantages experienced by the adjacent Woodshed Base or nearby Damper Public Wharf in terms of entry and departure draft restrictions.

The function and purpose of Berth 1 will be:

- To service the larger offshore supply boat market on a fast turnaround basis.
- To receive and offload very heavy ro/ro cargoes up to 1500 tonne delivered by ocean going heavy lift ships and barges.
- To handle inbound and outbound cargoes related to major offshore pipe lay projects.
- To receive and efficiently load reel ships used for deep water small diameter pipelay.

The wharf will be an earth filled structure with steel sheet pile faces and concrete capping beam surround. Most of the construction will be performed using land based equipment working from the core of the earth filled system.

Much effort has gone into a design concept which allows very large cranes (>100 tonne capacity) to operate without restriction on the wharf.

The separation between Berth 1 and Berth 2 is such to allow Road Train Triples (the max allowable) to turn unassisted on the wharf.

C. QUAY WALL (BERTH 2)

The inner berth, Berth 2 has a minimum depth alongside of 5.0 m allowing unrestricted operation of all the Mermaid fleet, and the majority of other vessels servicing the offshore oil/gas industry and mineral ports. This berth will offer excellent weather protection for small and medium size vessels.

D. BREAKWATER.

The rubble mount type breakwater will be an extension of the wharf, constructed using core and armor rock largely won from excavations on the Base. The excavations created will become depositories for dredge spoil.

Because the storm surge associated with major cyclones can be up to 7 m above chart datum (low tide), before imposing the wave height, a fully protective breakwater is not practical. The